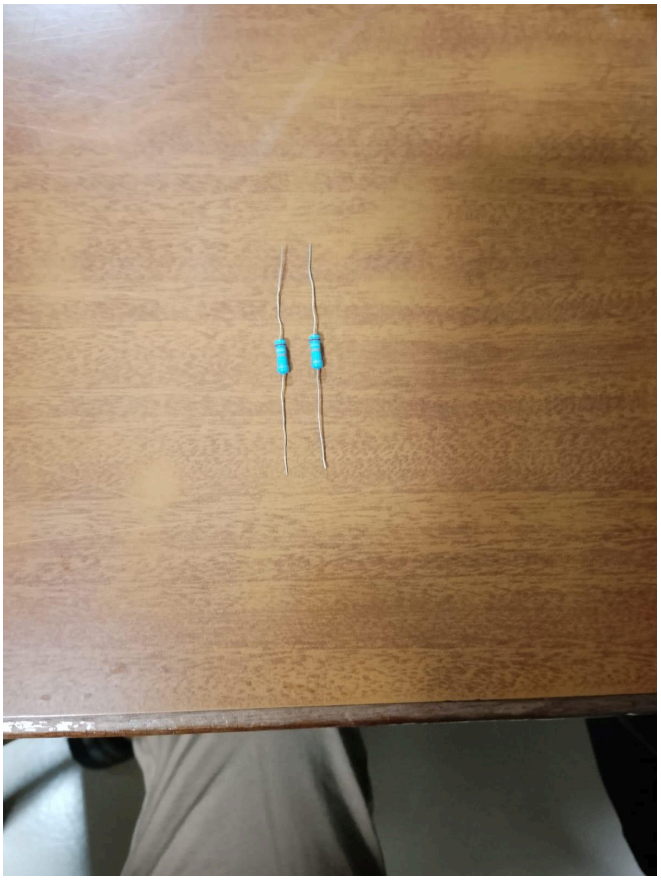
Practical Assignment 3
Experiment 1:
1. Pick one resistor, take a picture of it and determine its resistance value based on the colour code. Write down the resistance value.

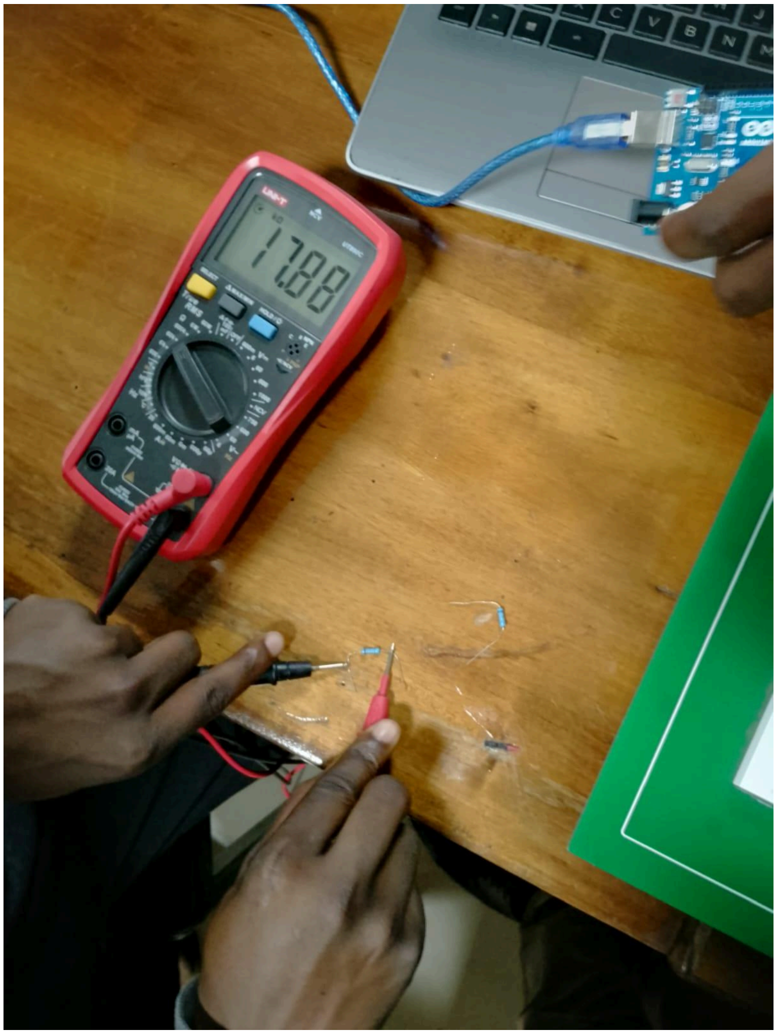


Colours: Brown, Grey, Orange ,Gold - 4 band

18 x 10^3 =18000
Gold = +- 5%
=18k ohms 5% tolerance

2. Using a multimeter, measure the resistance value emitted by the resistor. Take a

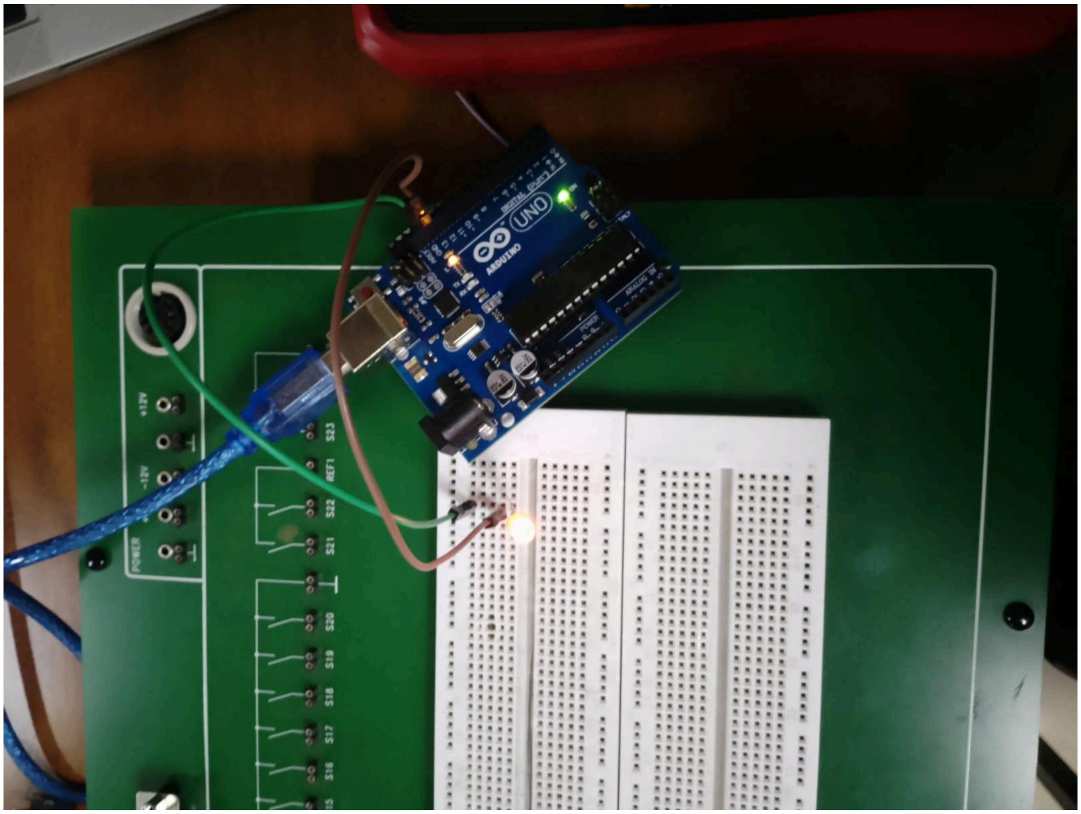
picture of this setup and write down the measured resistance.



Experiment 2:

1. Using the breadboard, microcontroller, and LED, recreate the following schematic.

Take a picture of the physical implementation.

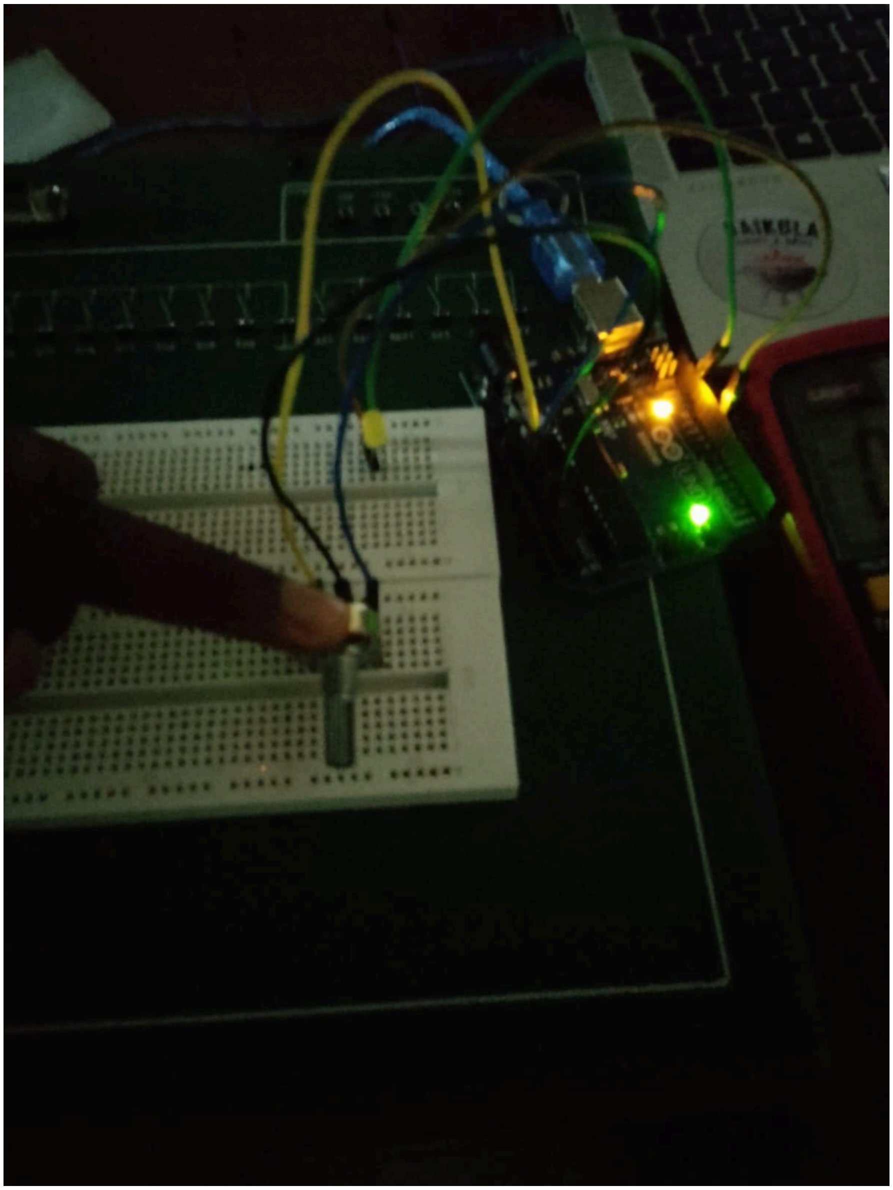


3. Thereafter, use the multimeter to determine the voltage passing through the circuit. Write down the value registered.

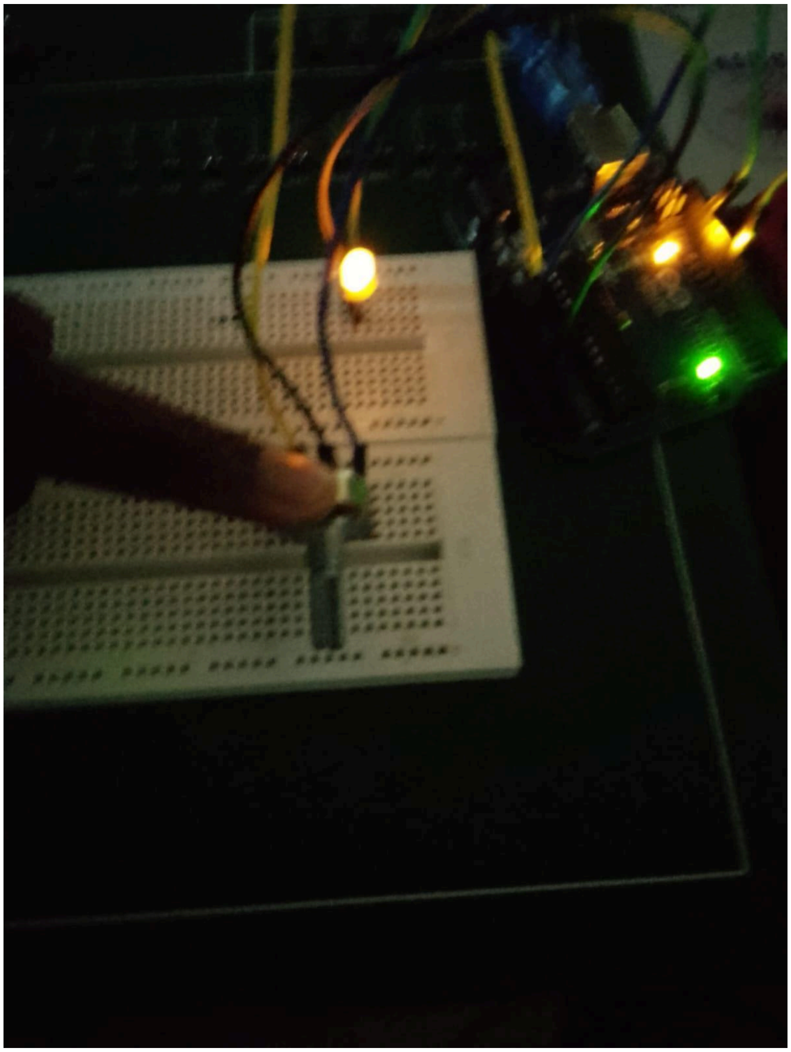
Values fluctuate between 0 - 0.8volts as the LED blinked

4. Reconfigure the circuit you created in question 1 by adding a potentiometer to simulate Pulse Width Modulation (PWM) on the LED. Take 3 images of your circuit with the LED at 25% brightness, 50% brightness and 100% brightness. You can refer to these or any other suitable resources to guide on the connection: Using PWM to Control the Light Intensity of a LED - Instructables; Arduino LED

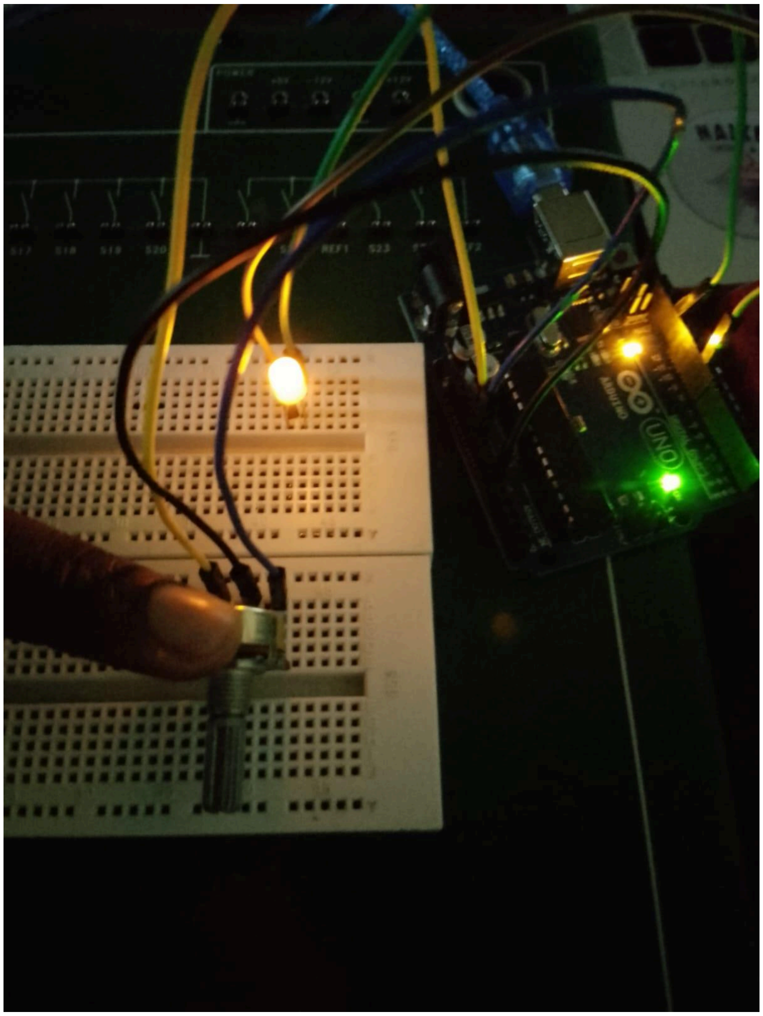
Dimmer (Potentiometer + PWM)



25% percent brightness



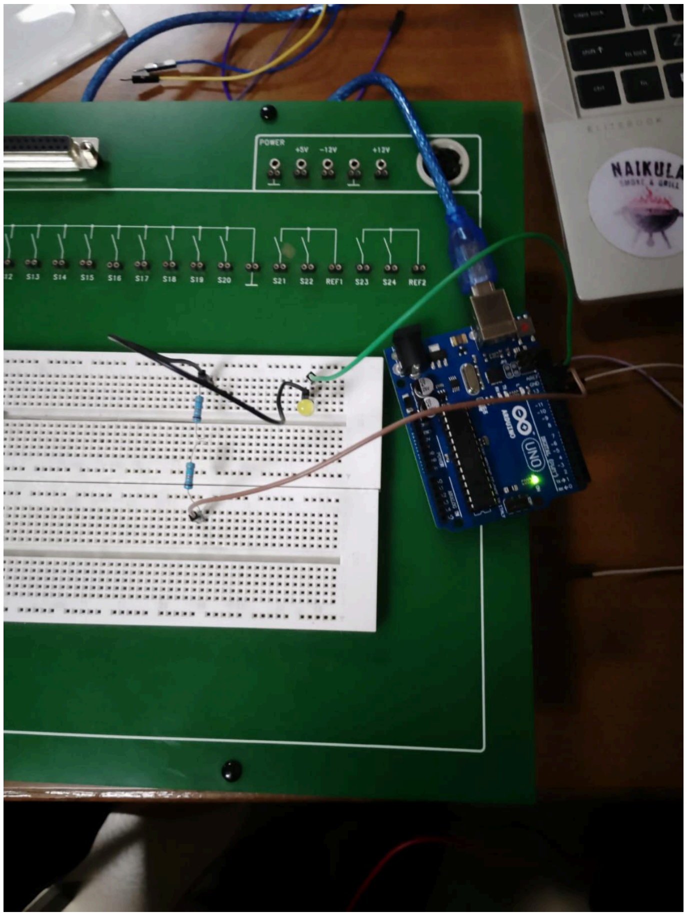
50% - brightness



100% - brightness

Experiment 3:

1. Using the breadboard, microcontroller, both resistors and LED, recreate the following schematic of resistors connected in series. Take a picture of the physical Implementation.

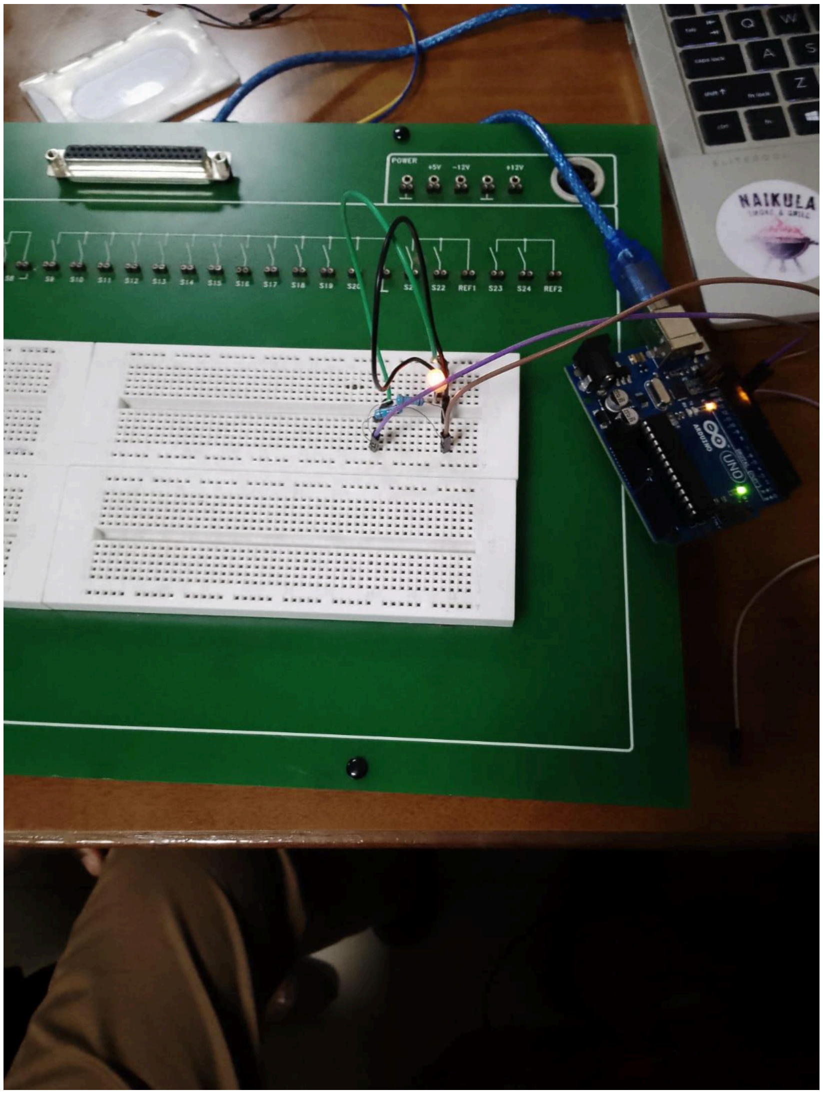


3. Thereafter, use the multimeter to determine the voltage passing through the circuit. Write down the value registered at each resistor.

Values change between 0 to approx. 0.569

Experiment 4:

1. Using the breadboard, microcontroller, both resistors and LED, recreate the following schematic of resistors connected in parallel. Take a picture of the physical Implementation.



3. Thereafter, use the multimeter to determine the voltage passing through the circuit. Write down the value registered at each resistor.

Each has 1 volt across each resistor

Group 10 participants